







# Random Variable Models and Simulation Concepts

## Unit B - Probability and Statistics

2024

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# Discrete and continuous random variable models

- **In Wikipedia** (https://en.wikipedia.org/wiki/List_of_probability_distributions): *"Many probability distributions that are important in theory or applications have been given specific names."*
- **Discrete random variables (DRVs)**: Binomial, Poisson, Bernoulli, geometric and negative binomial.
- **Continuous random variables (CRVs)**: Exponential, regular and uniform.
- **Indicators** are calculated from the parameters of each model.
  - Expectation → $E(X) = \mu_X$
  - Variance → $V(X) = \sigma_X^2$
- Probabilities are calculated from the **probability mass functions and cumulative distribution function**:

| DRV | CRV |
|---|---|
| $P(X=k) = p_X(k)$ | $P(X=k) = 0$ |
| $P(X \leq k) = F_X(k) = S_{j \leq k}\, p_X(j)$ | $P(X \leq k) = F_X(k)$ |
| $P(X<k) = P(X \leq k-1) = F_X(k-1)$ | $P(X<k) = P(X \leq k) = F_X(k)$ |
| $P(a<X \leq b) = F_X(b) - F_X(a)$ | $P(a \leq X \leq b) = F_X(b) - F_X(a)$ |

- In addition, inverse cumulative distribution functions can be calculated (given a probability α, calculate the **α quantile** or **α percentile in %**):

$x_\alpha$ is the α quantile of *X* if it holds that $F_X(x_\alpha) = \alpha$ $(0 \leq \alpha \leq 1)$

# Bernoulli model

- **Definition**: Number of successes when performing a single experiment with two possible outcomes*: **0** (“failure”) and **1** (“success”)

* Called binary or dichotomous responses.

- **Notation**: $X \sim \text{Bern}(p)$
- **Parameters**: $p$ (probability of success)
- **Probability mass function**:

| K | $P_X(k)$ |
|---|---|
| 0 | 1-p = q |
| 1 | p |

The values “0” and “1” can have a broad meaning:

- “1” means “success” in the option of interest. In a broad sense it can mean *right, positive, etc.*

- “0” means “failure” in the option of interest. In a broad sense it can mean *error*, *no success*, *negative*, etc.

- It is the simplest general theoretical model applicable to a random variable. The most common case is that of random experiments involving repeated Bernoulli trials. Some trials must be ***independent*** from others and the **probability of success must be constant** and equal to *p.*

# Models associated with Bernoulli trials

- In a random experiment involving repeated independent Bernoulli trials, the following **DRVs** are considered interesting distributions:
  - **Binomial:** total number of "successes" in n repetitions.
  - **Geometric:** number of repetitions until the first "success" is observed.
  - **Negative binomial:** number of repetitions until the *r*th "success" is observed.
- In random experiments where the number of repetitions *n* is very large and *p* is very small (*strange* phenomena), it may be easier to identify the mean (*np*) number of "successes" (in the interval unit) than the explicit value of *n* and *p*. In this case, the following are considered interesting distributions:
  - **Poisson:** number of "occurrences" in the interval → **DRVs**
  - **Exponential:** time between "occurrences" → **CRVs**

The latter are cases of a **Poisson point process** in which the Poisson DRV and the exponential CRV share a parameter or rate that relates the average number of "occurrences" and the time between "occurrences" (https://en.wikipedia.org/wiki/Poisson_point_process )

# Binomial model

- **Definition**: Number of successes in the repetition of *n* independent Bernoulli trials with constant probability *p*.
- **Notation**: $X \sim B(n,p)$

Functions in **R** for this model: dbinom(), pbinom(), qbinom()

- **Parameters**: number of repetitions (*n*); probability of observing one success (*p*).
- **Probability mass function**:

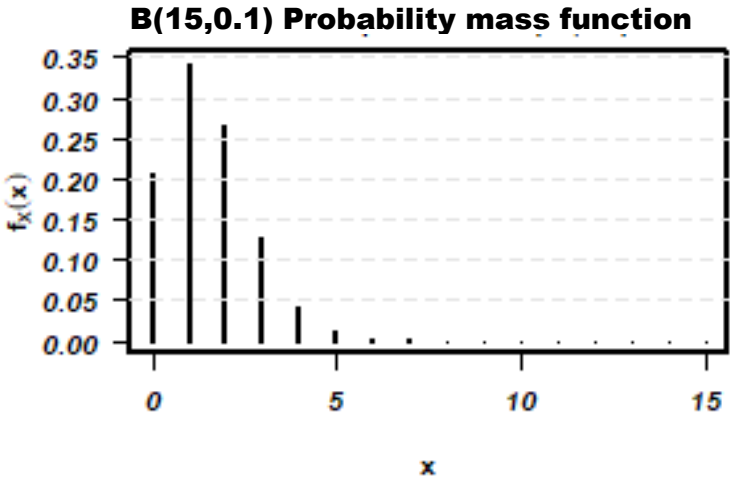
$$P(X = k) = \binom{n}{k} \cdot p^k \cdot q^{n-k} \quad with \quad k = 0, 1, \dots, n$$

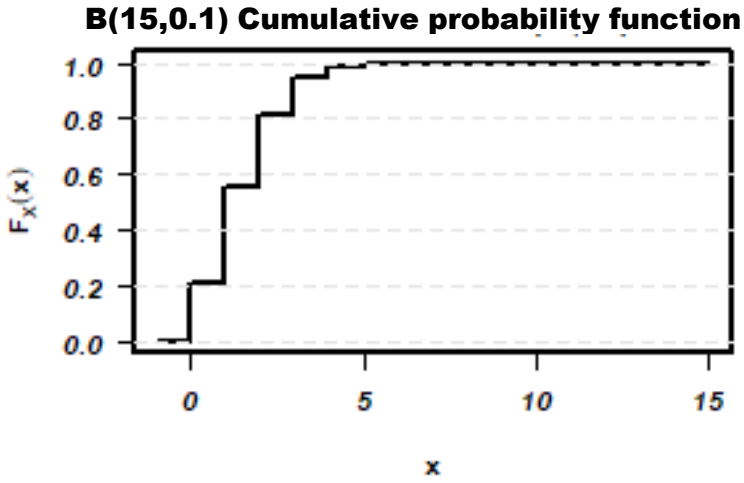
*where q = 1 – p and* $\binom{n}{k} = \frac{n!}{(n-k)!\,k!}$ *(o choose(n,k) in R, https://rdrr.io/snippets/ per R online)*

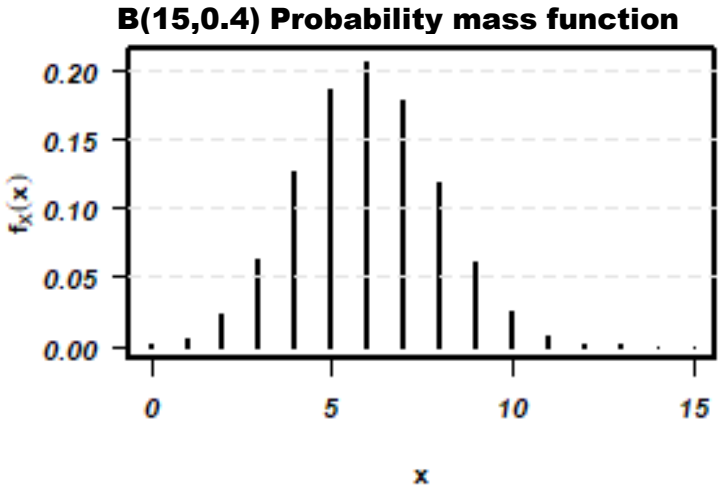
- It has no analytical **cumulative distribution function** → summation of point probabilities
- **Indicators:**
  - $E(X) = n \cdot p$
  - $V(X) = n \cdot p \cdot q$

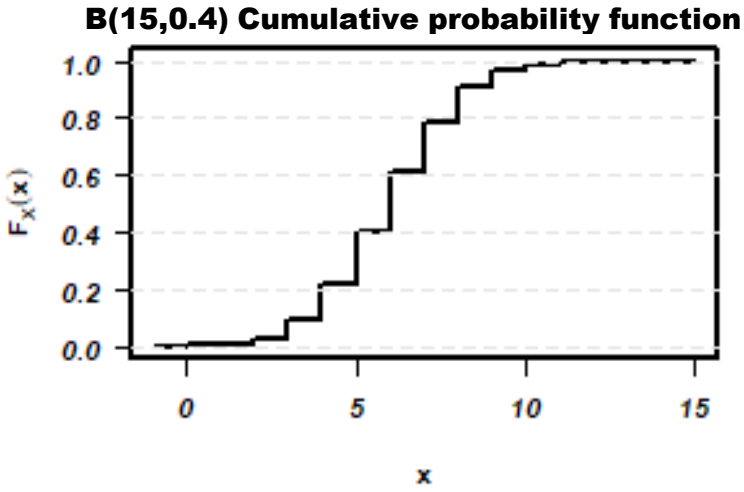
# Binomial model. Graphic representation

[E.g., How is the number of spam messages distributed among the first 15 received per day depending on whether the probability of a message being spam is 0.1 or 0.4?



# Binomial model. An example of probability calculation

Let $X$ follow the model B($n$=20,$p$=0.5)

- **Point probability**. What is the probability of X=14?
  - With formulas → $P(X = 14) = \binom{20}{14} \cdot 0.5^{14} \cdot 0.5^{6} = \mathbf{0.037}$
  - With R → $P(X = 14) = dbinom(x = 14, size = 20, prob = 0.5) = \mathbf{0.03696442}$

- **Cumulative probability**. What is the probability of X ≤ 14?
  - With formulas → $P(X \leq 14) = \binom{20}{0} \cdot 0.5^{0} \cdot 0.5^{20} + \cdots + \binom{20}{14} \cdot 0.5^{14} \cdot 0.5^{6} = \mathbf{0.979}$
  - With R → $P(X \leq 14) = pbinom(q = 14, size = 20, prob = 0.5) = \mathbf{0.9793053}$

- **Quantiles**. What is the value such that the probability of being below it is at least 0.95?
  - With formulas → $P(X \leq x_{0.95}) = 0.95 \rightarrow Very\ complicated$!!!
  - With R → $P(X \leq x_{0.95}) = 0.95 \rightarrow qbinom(p = 0.95, size = 20, prob = 0.5) = \mathbf{14}$

dbinom, pbinom, qbinom are functions in R, Rstudio, and R online (https://rdrr.io/snippets/)

Probabilities in Binomial model could be obtained using the model with parameter p or 1-p:
*P ( X ≤ k ) = 1 - P ( Y ≤ n-k-1 ) where X* ~ B(*n*, *p*) *and Y* ~B(*n*, 1-*p*) ( pbinom(k,n,p)=1-pbinom(n-k-1,n,1-p) )

# Geometric model

- **Definition**: number of attempts (k) of a Bernoulli trial until the **first success** is observed.
- **Notation**: $X \sim \text{Geom}(p)$

Functions in **R** for this model: dgeom(), pgeom(), qgeom()

The parameter in R functions is the number of failures (not number of attempts) *(k attempts are the same as k-1 failures)*

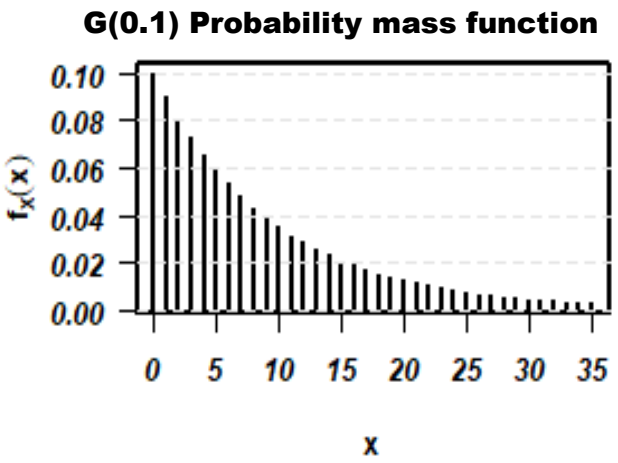
- **Parameters**: $p$ (probability of observing one success)
- **Probability mass function**:

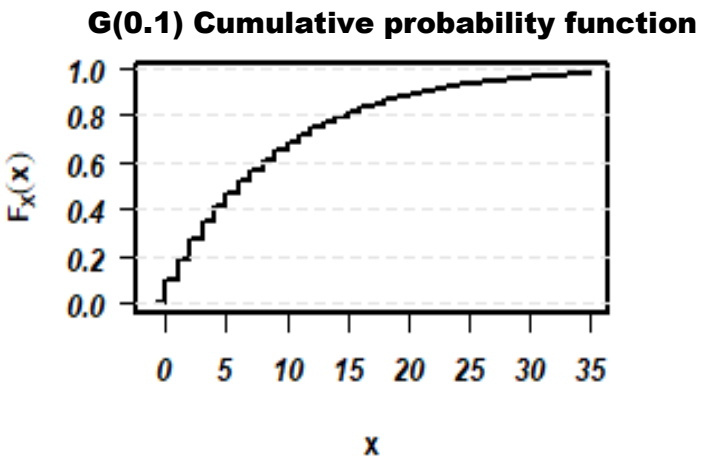
$$P(X = k) = q^{k-1} \cdot p \qquad k \geq 1$$

**Indicators:**

$E(X) = 1/p$

$V(X) = (1-p)/p^2$




It is not bounded; any integer value > 0 is possible. [E.g., "Roll the dice as many times until the first 1 comes up".] Although the number of attempts is most likely not very high, when $p$ increases, $P_X(k)$ is moved to lower values, and $F_X(k)$ grows faster.

# Negative binomial model

- **Definition**: number of attempts (*k*) of a Bernoulli experiment until *r* successes occur.
- **Notation**: $X \sim NB(r,p)$

Functions in **R** for this model: dnbinom(), pnbinom(), qnbinom()

The parameter in R functions is the number of failures (not number of attempts) *(k attempts are the same as k-1 failures)*

- **Parameters**: number of successes (*r*); probability to observe one success (*p*)
- **Probability mass function**:

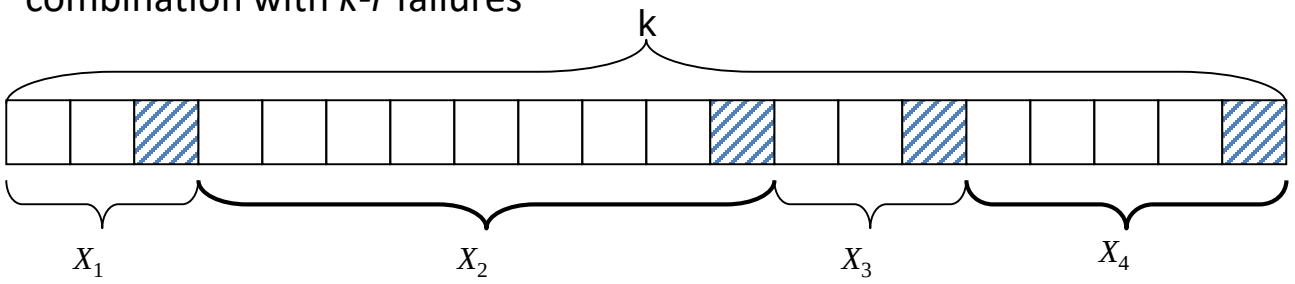
$$P(X = k) = \binom{k-1}{r-1} \cdot p^r \cdot q^{k-r}$$

**Indicators:**

$E(X) = r/p$

$V(X) = r(1-p)/p^2$

Not counting the last attempt (which has to be a success), *r* -1 successes are mixed in any combination with *k-r* failures




In general, we can consider a negative binomial as a sum of *r* independent geometries.
If *r*=1 is the geometric distribution

# Poisson model

- **Definition**: Number of occurrences in a given interval of time or space.

As in the binomial model, there may well be a repetition of identical Bernoulli-type experiments, but this model can also correspond to phenomena that occur unexpectedly (e.g., a call to a call center).

- **Notation**: $X \sim P(\lambda)$

Functions in **R** for this model: dpois(), ppois(), qpois()

- **Parameters**: $\lambda$ (event rate)
- **Probability mass function**: $$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \quad amb \quad k = 0, 1, \ldots$$

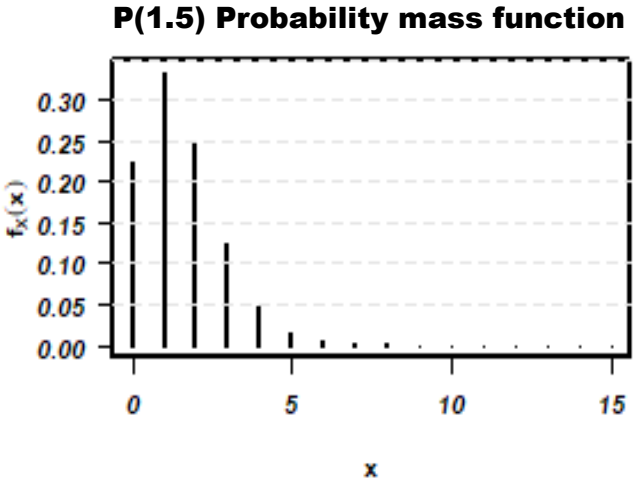
- **Indicators:**
  - $E(X) = \lambda$
  - $V(X) = \lambda$

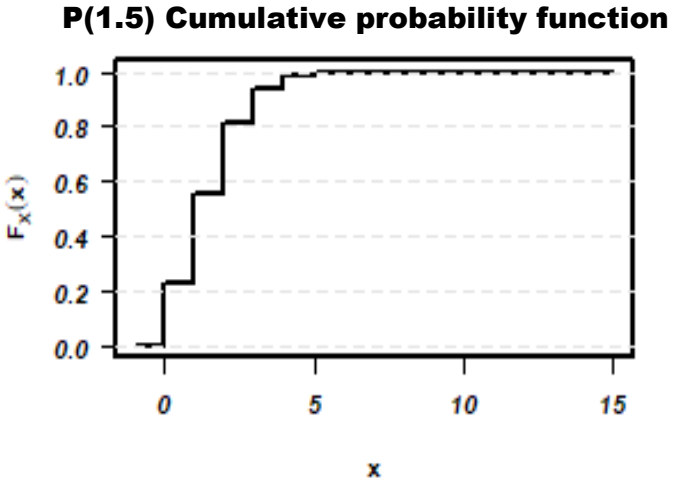
($\lambda$ is a positive real number that represents the average occurrence rate per time unit considered)

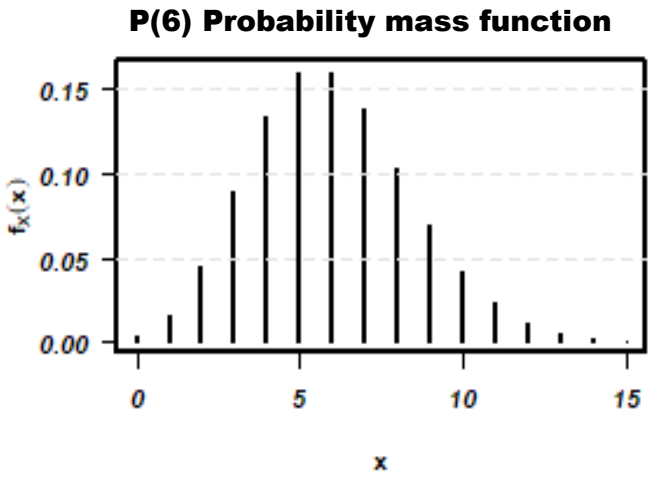
A Poisson variable can take any integer value $k \geq 0$, although in practice only those relatively close to $\lambda$ have non negligible probabilities.

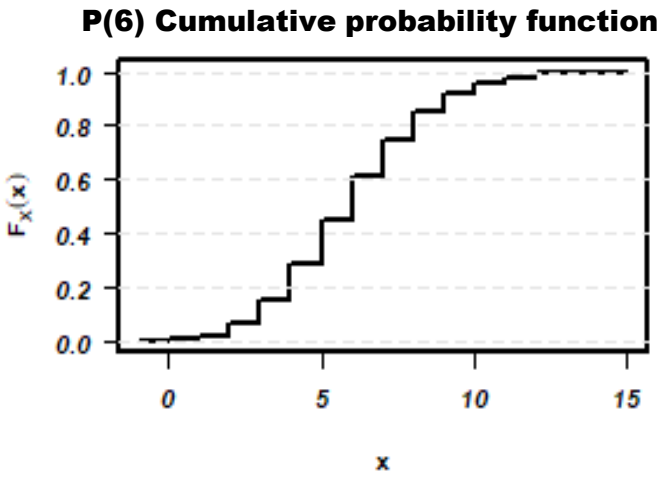
# Poisson model. Graphic representation

**EXAMPLE**. How is the number of spam messages received per day distributed depending on whether the value of the mean rate is 1.5 or 6?



# Poisson model. An example of probability calculation

Let $X$ follow the model P(λ =2):

- **Point probability**. What is the probability of X=3?
  - With formulas → $P(X=3)=\frac{e^{-2}\cdot 2^3}{3!}=\mathbf{0.1804}$
  - With R → $P(X=3)=dpois(x=3,\text{lambda}=2)=\mathbf{0.1804470}$
- **Cumulative probability**. What is the probability of X ≤ 3 ?
  - With formulas → $P(X\leq 3)=\frac{e^{-2}\cdot 2^0}{0!}+\cdots+\frac{e^{-2}\cdot 2^3}{3!}=\mathbf{0.857}$
  - With R → $P(X\leq 3)=ppois(q=3,\text{lambda}=2)=\mathbf{0.8571235}$
- **Quantiles**. What is the value such that the probability of being below it is at least 0.95?
  - With formulas → $P(X\leq x_{0.95})=0.95\rightarrow Very\ complicated$!!!
  - With R → $P(X\leq x_{0.95})=0.95\rightarrow qpois(p=0.95,lambda=2)=\mathbf{5}$

dpois, ppois, qpois are funcions in R, Rstudio, and R online (https://rdrr.io/snippets/)

The sum of Poisson DRVs is a Poisson DRV with parameter $\lambda$ equal to the sum of the parameters $\lambda$'s of each one:

X ~ P(λ1) Y ~ P(λ2) → X+Y ~ P(λ1+ λ2)

From a Poisson DRV, we can define another by applying the change in the interval proportionally to the parameter:

*X="...in interval t"* ~ P(λ) → *Y="...in interval kt"* ~ P(kλ)

# Summary of DRV models

| Distribution | Notation | Domain | Expectation $E(X) = \mu_X$ | Variance $V(X) = \sigma_X^2$ |
|---|---|---|---|---|
| **Bernoulli** | Bern(p) | 0, 1 | p | p·q |
| **Binomial** | B(n,p) | 0,1,...,n | n·p | n·p·q |
| **Geometric** | Geom(p) | 1.2,3,... | 1/p | $q/p^2$ |
| **Negative binomial** | BN(r,p) | r, r+1,... | r/p | $q \cdot r/p^2$ |
| **Poisson** | P(λ) | 0, 1, 2,... | λ | λ |

$0 < p < 1$ $\quad q = 1 - p$ $\quad n \in N$ $\quad r \in N$ $\quad \lambda \in R^+$

# Exponential model

- **Definition**: Time distribution between events (occurrences) in a Poisson process. That is, if in the interval [0, t] the events to the system ($N_t$) follow a Poisson distribution, with the rate $\lambda \cdot t$ (the rate per unit of time is $\lambda$), then the time between two consecutive events is a continuous and non deterministic quantity that is exponentially distributed. [E.g., the lifetime of an electronic component.]




- **Notation**: $X \sim \text{Exp}(\lambda)$

Functions in **R** for this model: dexp(), pexp(), qexp()

- **Parameters**: $\lambda$ (event occurrence rate per unit time)
- **Density and Cumulative distribution function**:

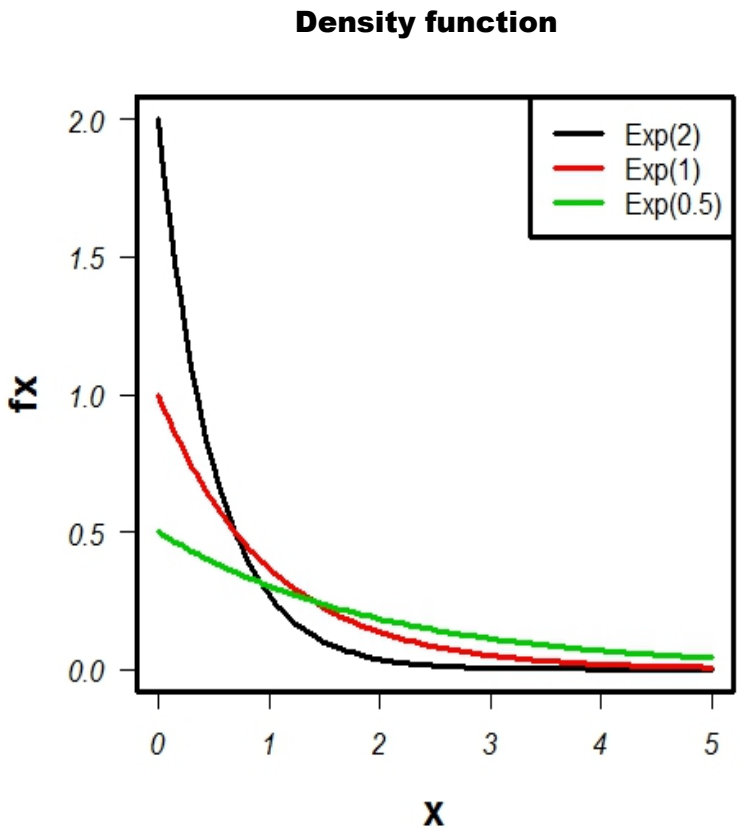
$$f_X(x) = \lambda \cdot e^{-\lambda x} \qquad with \quad x > 0$$

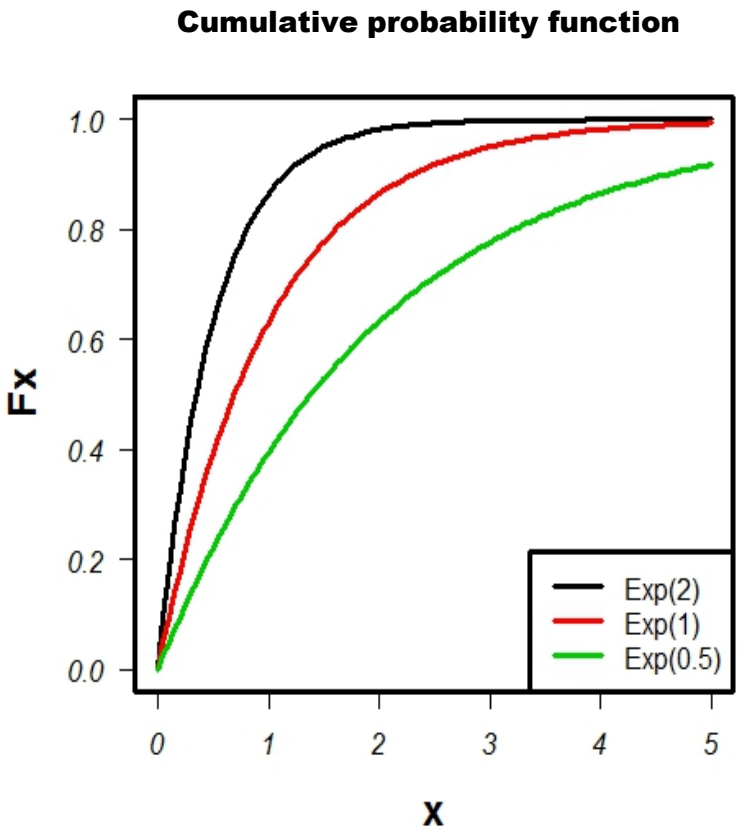
$$F_X(x) = 1 - e^{-\lambda x} \qquad with \quad x > 0$$

- **Indicators:**
  - $E(X) = 1/\lambda$
  - $V(X) = 1/\lambda^2$

# Exponential model. Graphic representation

[E.g., How is time between spam messages distributed if I get two spam mails per hour in average? What if I get one per hour? What if I get one every two hours?



# Exponential model. An example of probability calculation

Let $X$ follow the model Exp($\lambda$=2):

- **Point probability**. → Remember that $P(X=x) = 0$ for any x because it is a CRV.

- **Cumulative probability**. What is the probability of X ≤ 2?
  - With formulas → $P(X \leq 2) = 1 - e^{-2 \cdot 2} = 1 - e^{-4} = \mathbf{0.9817}$
  - With R → $P(X \leq 2) = pexp(q = 2, \text{rate} = 2) = \mathbf{0.9816844}$

- **Quantiles**. What is the value such that the probability of being below it is at least 0.95?
  - With formulas → $P(X \leq x_{0.95}) = 0.95 \rightarrow 1 - e^{-2 \cdot x_{0.95}} = 0.95 \rightarrow x_{0.95} = \mathbf{1.4979}$
  - With R → $P(X \leq x_{0.95}) = 0.95 \rightarrow qexp(p = 0.95, rate = 2) = \mathbf{1.497866}$

pexp, qexp are funcions in R, Rstudio, and R online (https://rdrr.io/snippets/)

*The exponential distribution has a distribution function with an explicit analytical expression.*

# Exponential model (some features)

- $f_X(x)$ is not equal to $P(X=x)$ [$P(X=x) = 0$ by definition] → $f_X(x)$ is not a probability, unlike the $p_X(x)$ of DRVs.
- Remember that in a CRV

$$P(a \leq X) = P(a < X) \quad \text{and} \quad P(a \leq X \leq b) = P(a < X < b) = F_X(b) - F_X(a).$$

Cumulative probabilities in the exponential model are calculated directly with the probability distribution function: $F_X(x) = 1 - e^{-\lambda \cdot x}$.

- **Markov (or memoryless) property:** the probability distribution of an exponential random variable does not depend on what happened before the current moment. Mathematically,

$$P(T > t_1 | T > t_0) = P(T > t_1 - t_0) \quad for\ t_1 > t_0$$

$P(T>t_1 | T>t_0) \neq P(T>t_1)$

- E.g., At a given moment, no requests have arrived for 10'' on a database server. Which is more likely: (A) to receive a request in the next 10'', or (B) to receive a request 10'' after another one?

  Solution: The probability is the same.

# Uniform model (continuous)

- **Definition**: CRV with a constant density function in a certain range. [The probability of belonging to a particular interval in that range depends only on the interval length]
- **Notation**: $X \sim U(a,b)$

Functions in **R** for this model: dunif(), punif(), qunif()

- **Parameters**: minimum value of the range of $X$ ($a$); maximum value of the range of $X$ ($b$).
- **Density and distribution function**:

**Constant!!!** → $f_X(x) = \frac{1}{b-a} \quad with \quad a < x < b$

**$F_X(x) = 0$ si $x < a$**
**$F_X(x) = 1$ si $x > b$**

→ $F_X(x) = \frac{x-a}{b-a} \quad with \quad a < x < b$

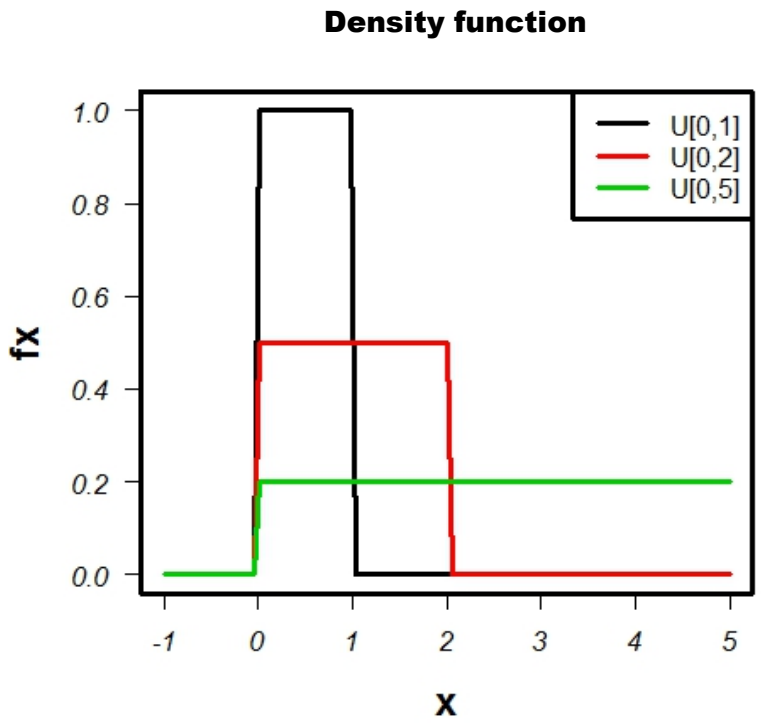
- **Indicators:**
  - $E(X) = (b + a)/2$
  - $V(X) = (b - a)^2/12$

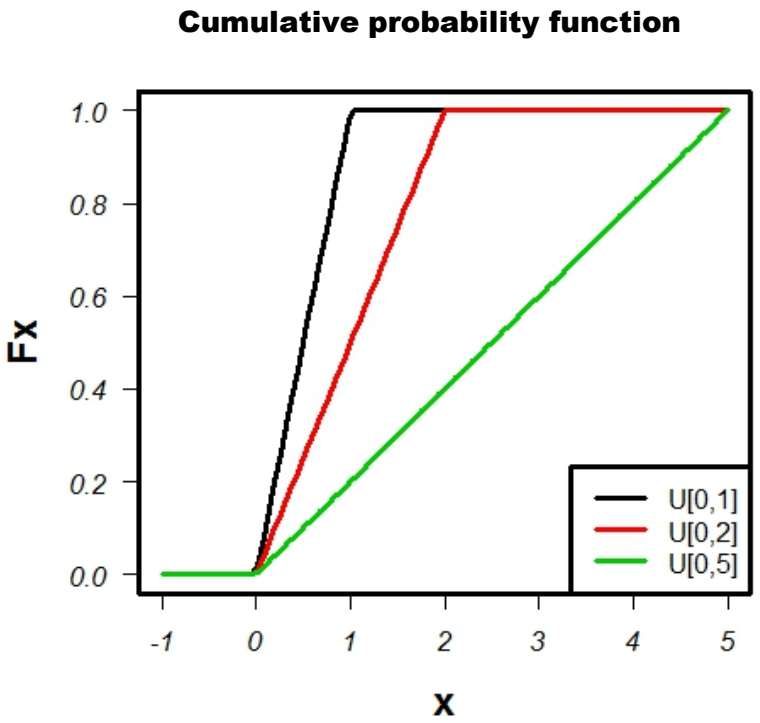
$$E(X) = \mu_X = \int_{-\infty}^{+\infty} x \cdot f_X(x)dx = \int_a^b x \cdot \frac{1}{b-a}dx =$$
$$= \frac{1}{b-a} \cdot \frac{x^2}{2}\Big]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{b+a}{2}$$

Intuitively, you can already see that the expected mean must be the centre of the interval

# Uniform model. Graphic representation

E.g., How are random real numbers between 0 and 1 distributed? And between 0 and 2? And between 0 and 5?



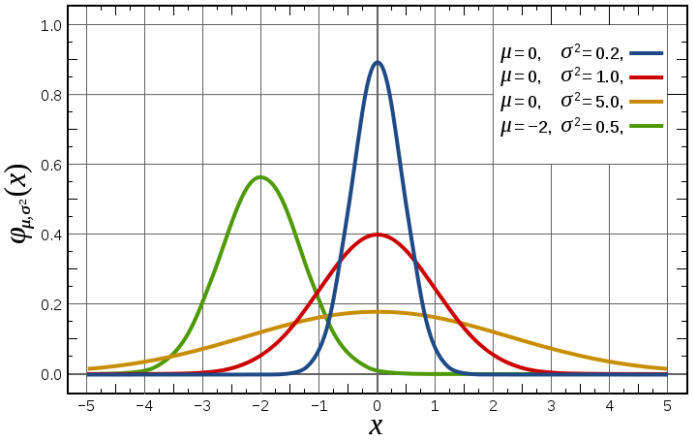



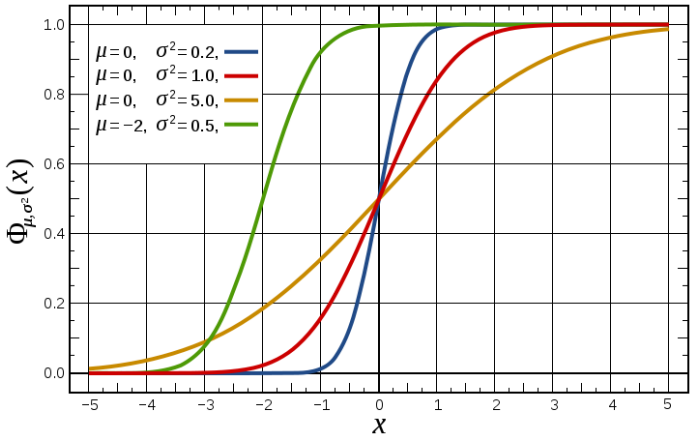
# Normal (or Gaussian) model. Introduction

(Wikipedia.org) *Normal Distribution:*

- "... the **normal** (or **Gaussian**) **distribution**, is a continuous probability distribution that is often used as a first approximation to describe real-valued random variables that tend to cluster around a single mean value."

- "... the normal distribution is **commonly encountered in practice**, and is used throughout statistics, natural sciences, and social sciences..."



# Normal model

- **Definition**: A model that serves to represent values from many phenomena that are found in scientific disciplines. [E.g., heights of people, the effect of a treatment, noise in telecommunications.]
- **Notation**: $X \sim N(\mu,\sigma)$

Functions in **R** for this model: dnorm(), pnomr(), qnorm()

- **Parameters**: expectation ($\mu$ ); standard deviation ($\sigma$)

[$\sigma^2$ is sometimes used as a parameter instead of σ]

- **Density function**:

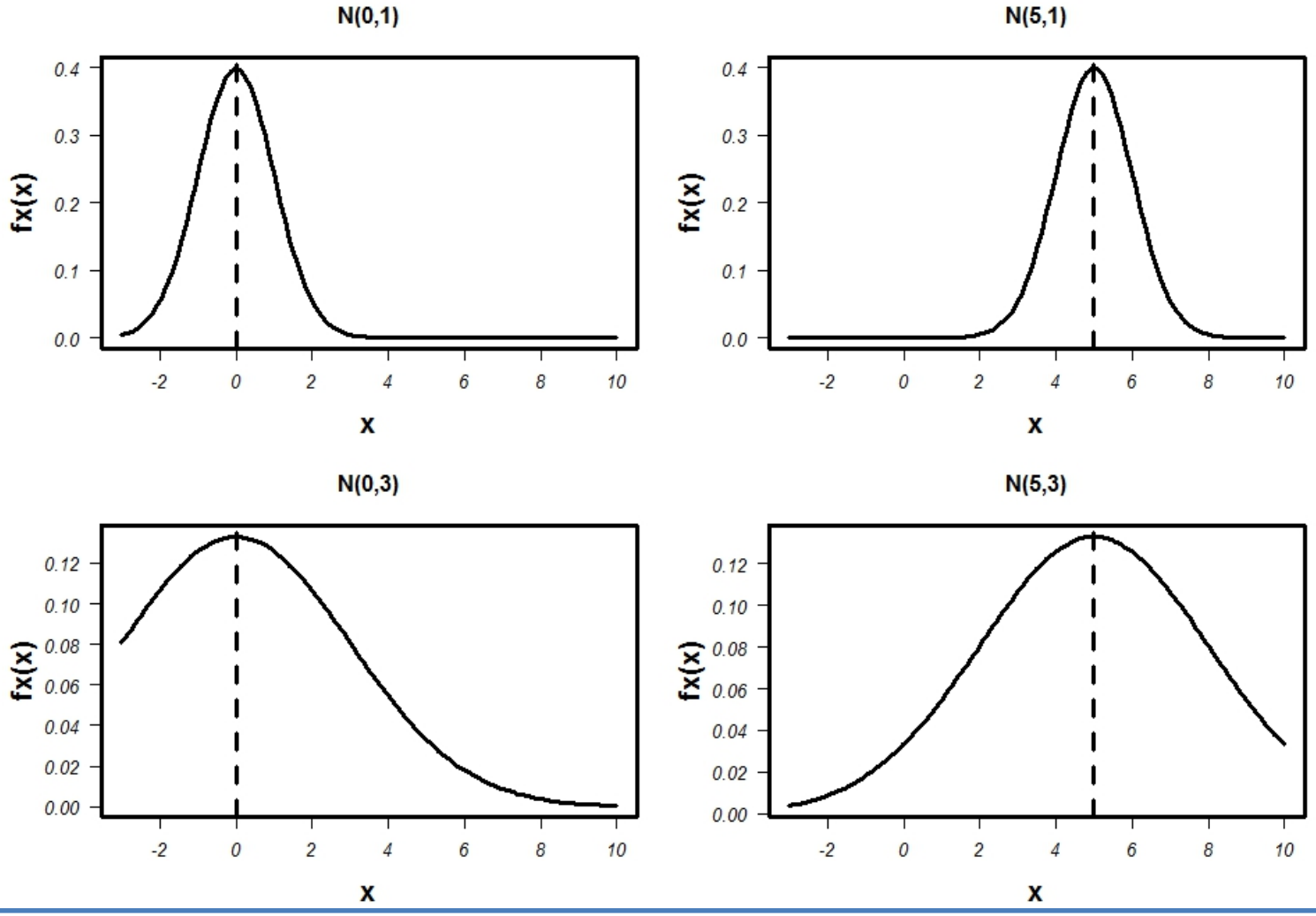
$$f_x(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad with \quad x \in R$$

- The **distribution function** has no explicit analytical expression → R
- **Indicators:**
  - E(X) = μ
  - V(X) = $\sigma^2$

The normal distribution with parameters $\mu$ = 0 and $\sigma$=1 is called the **standard normal distribution**

# Normal model. Graphic representation

E.g., The density functions of different normal distributions according to the values of $\mu$ and $\sigma$.



# Normal model. An example of probability calculation

Let $X$ follow the model N($\mu$=0, $\sigma$=1):

- **Point probability**. → Remember that $P(X=x) = 0$ for any $x$ because it is a CRV.
- **Cumulative probability**. What is the probability of $X \leq 2$?
  - With formulas → **It can't be done!!!**
  - With R → $P(X \leq 2) = pnorm(q = 2, \text{mean} = 0, sd = 1) = \mathbf{0.9772499}$
- **Quantiles**. What is the value such that the probability of being below it is at least 0.95?
  - With formulas → **It can't be done!!!**
  - With R → $P(X \leq x_{0.95}) = 0.95 \rightarrow qnorm(p = 0.95, mean = 0, sd = 1) = \mathbf{1.645}$

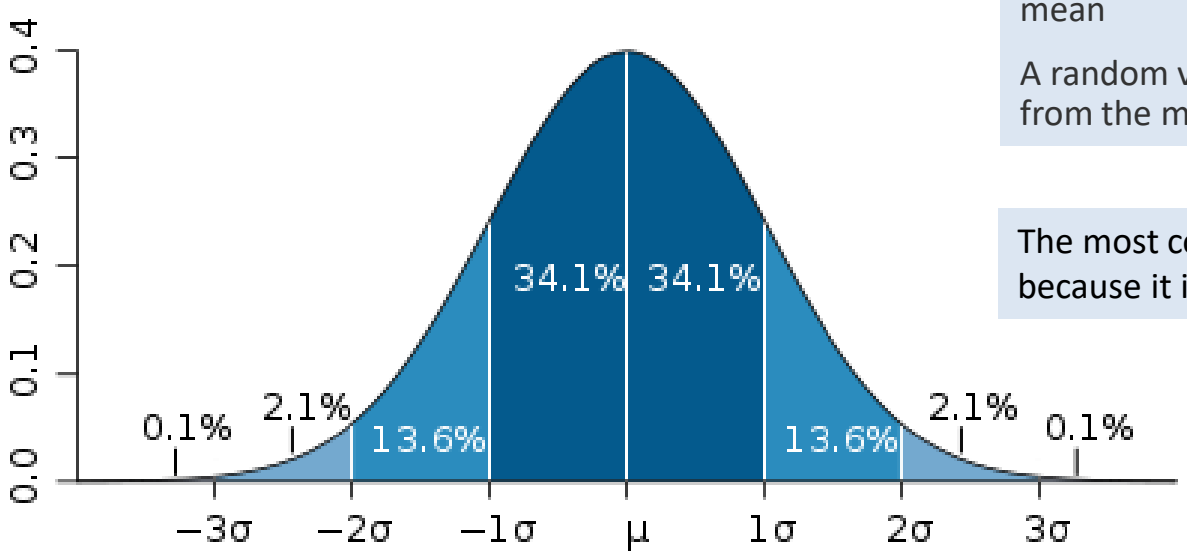
$$(qnorm(0.95) = 1.64)$$

pnorm and qnorm are funcions in R, Rstudio, and R online (https://rdrr.io/snippets/)

R functions for the Normal model can be used for all values of the parameters, but the default values, $\mu$ = 0 and $\sigma$=1, corresponds to the **standard normal distribution.** In next slides we will see a standardisation process

# Normal model. Properties and quantiles

- The density function $f(x)$ is symmetric about the point $x = \mu$, which is both the mean and the median of the distribution.
- The inflection points are at one standard deviation from the $\mu$ ($x=\mu-\sigma$ and $x=\mu+\sigma$).
- The quantiles of the standard normal distribution $Z \sim N(0,1)$ are usually denoted by $z_p$. The quantile $z_p$ represents a value such that in a standard normal distribution it has a probability $p$ of falling in the interval $(-\infty, z_p)$.




In practice, $X$ has more density close to the mean

A random value will not be further than **2σ** from the mean μ, with a **95.4%** probability

The most common quantile is $\boldsymbol{z_{0.975} = 1.96}$ because it is widely used in statistical inference.

# Normal model. Standardisation

- A **linear combination of normal variables** is normal:
  - Let $a$ and $b$ be two real numbers, and $X \sim N(\mu_X, \sigma_X) \rightarrow Y = a \cdot X + b \sim N(\boldsymbol{\mu_Y} = a \cdot \mu_X + b, \boldsymbol{\sigma_Y} = a \cdot \sigma_1)$
  - Let $a$ and $b$ be two rela numbers, $X \sim N(\mu_X, \sigma_X)$ and $Y \sim N(\mu_Y, \sigma_Y) \rightarrow$

    $\rightarrow W = a \cdot X + b \cdot Y \sim N\left(\boldsymbol{\mu_W} = a \cdot \mu_X + b \cdot \mu_Y, \boldsymbol{\sigma_W} = \sqrt{a^2 \sigma_1^2 + b^2 \sigma_2^2 + 2 \cdot ab \cdot \rho_{XY} \cdot \sigma_1 \cdot \sigma_2}\right)$.

- This property can be used to relate normal distributions through translations and scale changes. In particular, transforming to the standard normal $Z \sim N(0,1)$, **standardising**, allows you to search the tables of $Z$ for probabilities of any normal distribution.
- With $X \sim N(\mu, \sigma)$ and $Z \sim N(0, 1)$, we can relate:

  $Z = X/\sigma - \mu/\sigma = (X - \mu) / \sigma \sim N(0, 1)$ ($a = 1/\sigma$, $b = -\mu/\sigma$ are scalars). In other words,

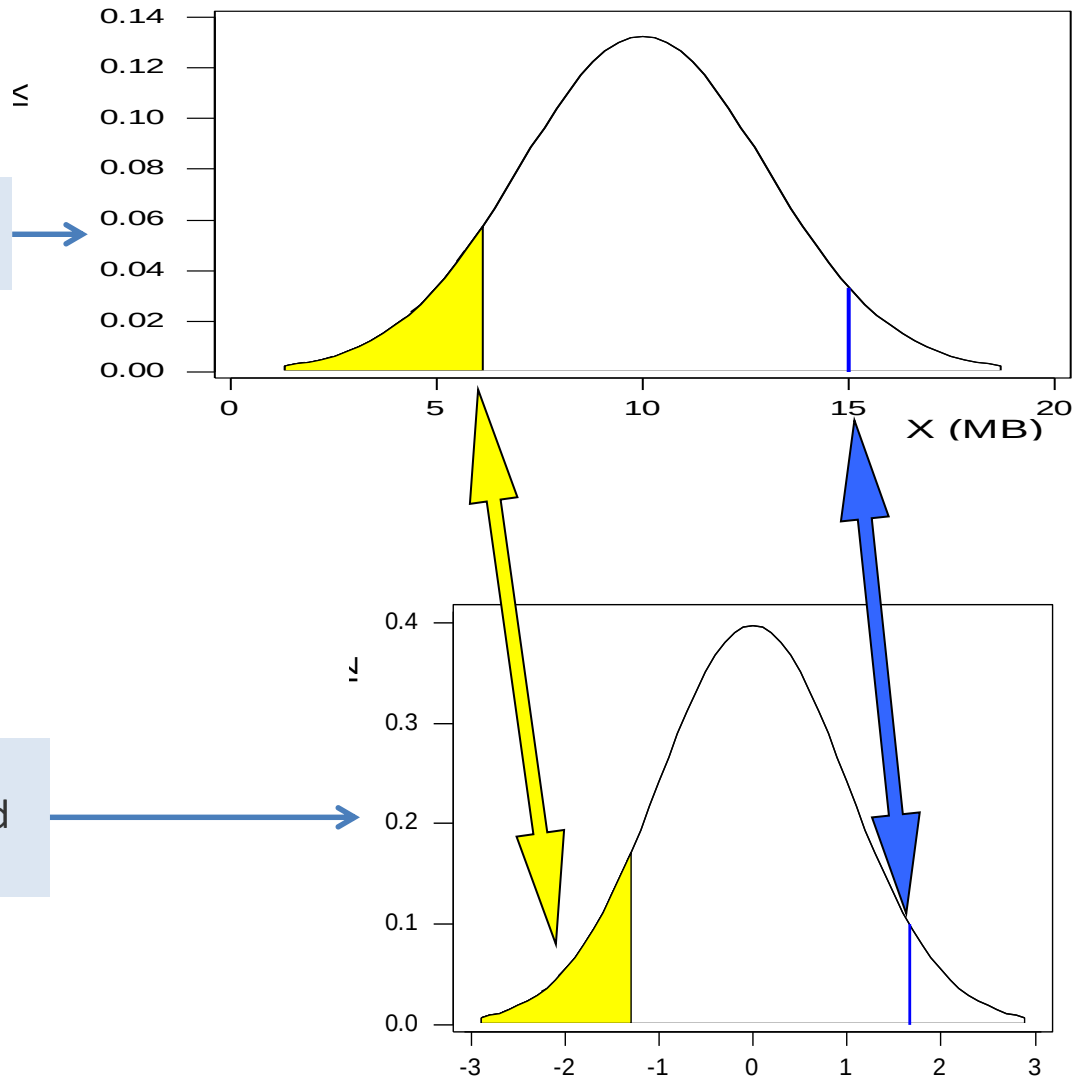
$$\boldsymbol{Z = \frac{X - \mu}{\sigma} \sim N(0, 1) \rightarrow X = \mu + \sigma \cdot Z}$$

# Normal model. Standardisation

**Variable *X***: actual distribution (e.g., Mb of a file).

$$Z = \frac{X - \mu}{\sigma}$$

**Variable *Z***: standardised distribution, unitless, centered at 0, with standard deviation equal to 1

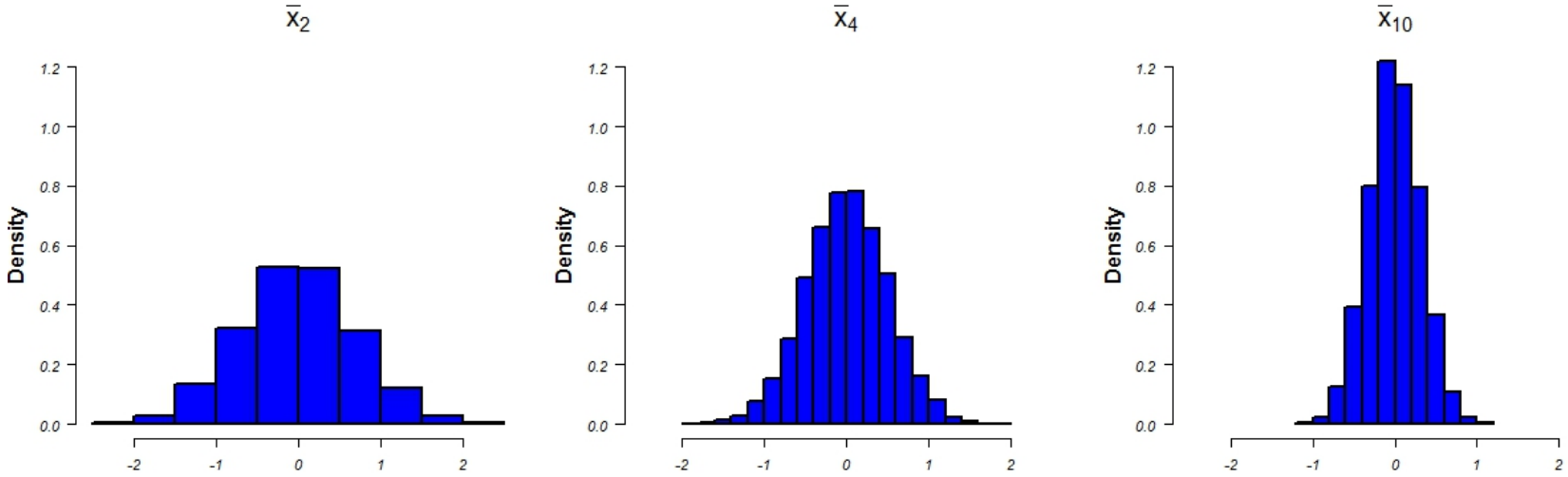


# Distribution of the mean of RVs

- We have simulated $\bar{X} = (X_1 + X_2 + \cdots + X_n)/n$ such that $X_i$ are IID. We observe that
  - **it tends to concentrate around μ** when *n* increases,
  - **it tends to resemble a normal distribution** as *n* becomes large.

$$E(\bar{X}_n) = \frac{E(\sum X_i)}{n} = \frac{\sum E(X_i)}{n} = \frac{n\mu}{n} = \mu \qquad V(\bar{X}_n) = \frac{V(\sum X_i)}{n^2} = \frac{\sum V(X_i)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

- For any *n*, the expectation of the sample mean is $\mu$ and the variance decreases with *n*: with a large sample, using the sample mean we get closer to μ.



# Central limit theorem (CLT)

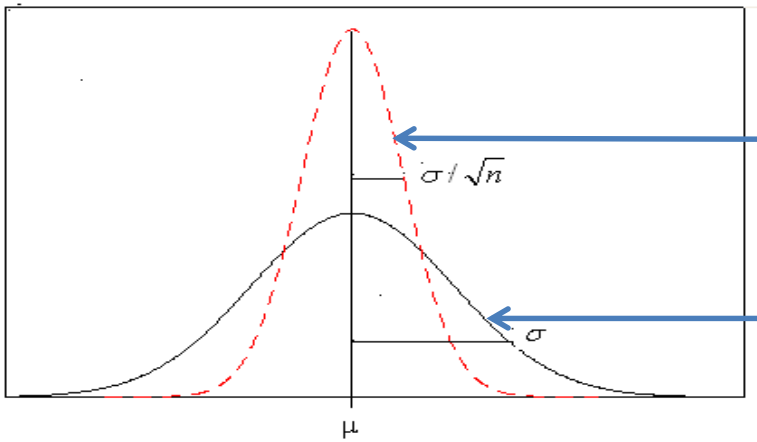
- Let $X_1$, $X_2$, …, $X_n$ be independent and identically distributed with expectation $\mu$ and standard deviation $\sigma$. Then,

$$S_n = \sum X_i \xrightarrow[large\ n]{} N(n\mu, \sigma\sqrt{n}) \Rightarrow \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow[large\ n]{} N(0,1)$$

$$\bar{X}_n = \frac{\sum X_i}{n} \xrightarrow[large\ n]{} N(\mu, \sigma/\sqrt{n}) \Rightarrow \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow[large\ n]{} N(0,1)$$

- I.e., with large *n*, the distribution function of the variable sum ($S_n$) and sample mean ($\bar{X}_n$) tends to a normal distribution with certain parameters **regardless of the distribution of the $X_i$**!
- See app.

# Central limit theorem (CLT)




$$\bar{X}_n = \frac{\sum X_i}{n} \xrightarrow[large\ n]{} N(\mu, \sigma/\sqrt{n})$$

$X_i$

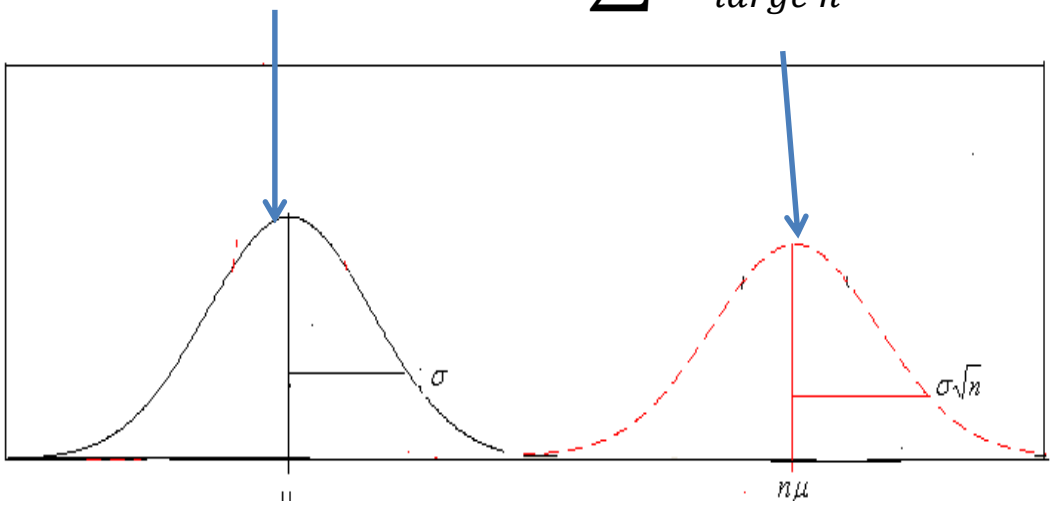
$$S_n = \sum X_i \xrightarrow[large\ n]{} N(n\mu, \sigma\sqrt{n})$$

**The $X_i$ do not necessarily have to be normal!!!!**

**They only have to comply with:**

**E(X) = μ**

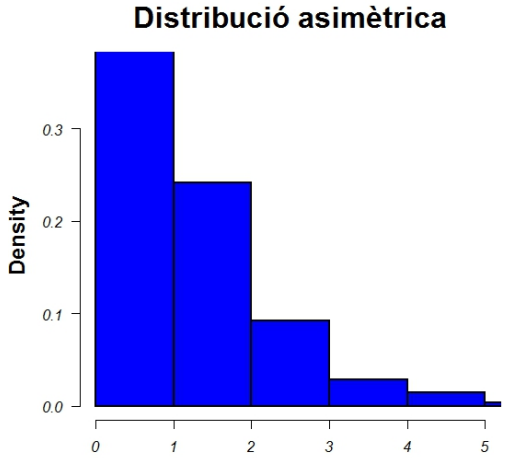
**V(X) = σ²**

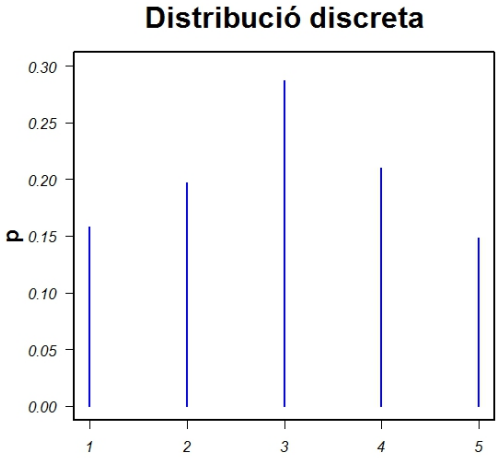


# Central limit theorem (CLT). "n"

## When is n *large enough* to apply the CLT?

- It depends on the original distribution and what you want to calculate.
- Convergence to normal is slower if the distribution of the $X_i$ is **not symmetric** or the **variables are discrete** (especially if they can only take a few values):




- Applications of the CLT: the normal curve approximates certain distributions well. [E.g., Poisson random variable, if $\lambda$ is large. The Student's t and the $\chi^2$ distribution are derivatives of normal distribution that will be seen below.]

# CLT. Relationships between distributions

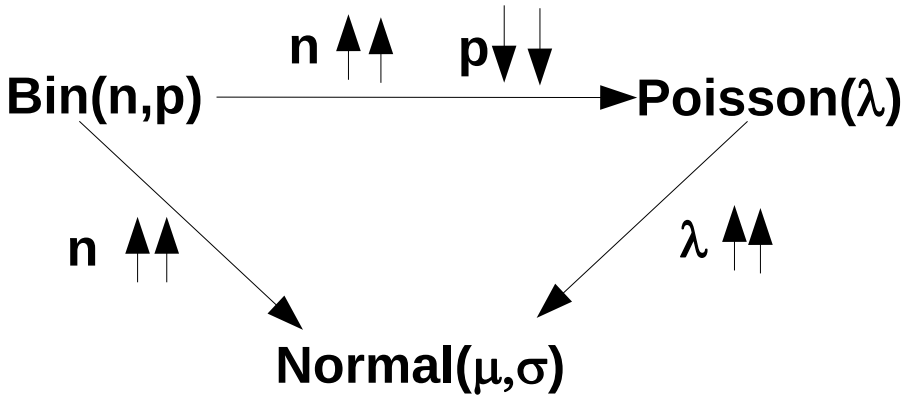
One of the practical applications of the CLT is that normal distribution can be used as an approximation of other distributions:

- The **binomial distribution** (Sum of Bernoulli RVs) with parameters *n* and *p* can be approximated by a normal distribution when *n* is large and *p* not too extreme (not very close to 0 or 1). Then, the parameters of the normal distribution are

  $\mu = n \cdot p$ and $\sigma^2 = n \cdot p \cdot (1-p)$

- The **Poisson distribution** with parameter $\lambda$ can be approximated by a normal distribution when the $\lambda$ is large enough Then, the parameters of the normal distribution are

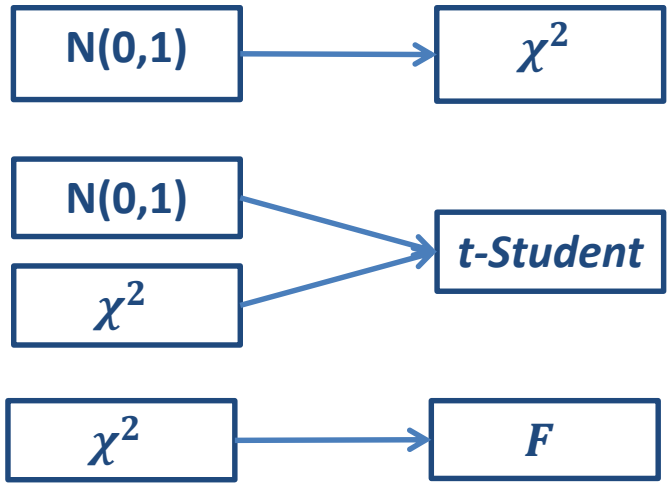
  $\mu = \lambda$ and $\sigma^2 = \lambda$




**N.B.**: The binomial distribution can be approximated to a Poisson distribution when *n* is large enough and *p* is small enough. Then $\lambda = np$

# Models derived from the Normal: $\chi^2$ *t-Student* *F*

- Three new distributions will be used in the next units for inference: $\chi^2$ and **t-Student** and ***F***
- These distributions are combinations of RV from other distributions:




- Next units of inference will describe these distributions to model statistics (calculated with data) instead of using them to describe real-valued random variables that tend to cluster around a single mean value

# Model derived from the Normal: $\chi^2$

- **Definition**: Let $X_i$ follow N(0,1) then:

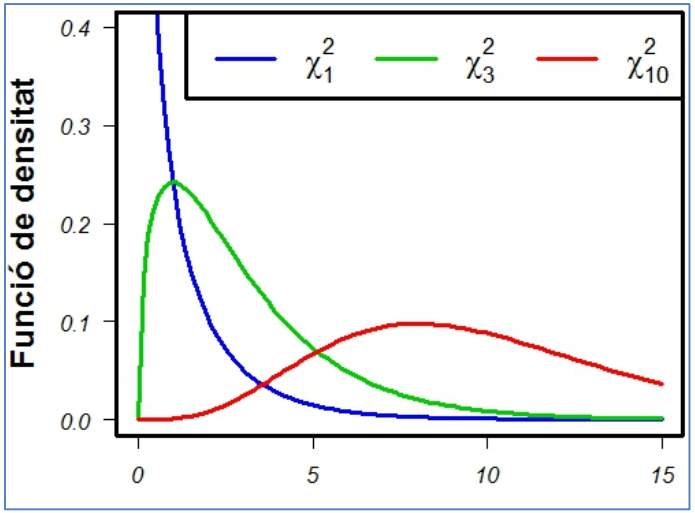
$$X_1^2 + X_2^2 + \cdots + X_n^2 \sim \chi_n^2$$

[for n = 1 $\rightarrow$ $X_1^2 \sim \chi_1^2$]

- **Notation**: $X \sim \chi_n^2$
- **Parameters**: *n* (degrees of freedom)
- **Density and Cumulative distribution function** :

$$f(x) = \frac{x^{k/2-1} \cdot e^{-x/2}}{2^{k/2} \cdot \Gamma(k/2)} \quad for \;\; x > 0$$

$$F(x) = \frac{\gamma(k/2, x/2)}{\Gamma(k/2)} \quad for \;\; x > 0$$




*Γ* : Gamma function
*γ* : Incomplete Gamma function
*n*: degrees of freedom

**R**: dchisq, pchisq, qchisq

*Script showing that the sum of squared Normals is $\chi^2$*

```
M = 500                                          # Samples of Normals
n = 7                                            # Degrees of freedom
sample = array(rnorm(M*n),dim=c(M,n))            # n samples of N(0,1)
sample2 = sample*sample                          # n samples of (N(0,1))^2
sum = apply(sample2, 1, sum)                     # Sum of squared samples
hist(sum, breaks="Scott", freq=FALSE)            # Empirical Distribution of "sum"
curve(dchisq(x, n),add=TRUE,col=2,lwd=2)# Theoretical chi-square distribution
quantile(sum,c(0.25,0.50,0.75))                  # Q1, Median and Q3 of "sum"
qchisq(c(0.25, 0.50, 0.75),n)                    # Q1, Median and Q3 of chi-square
```

# Model derived from the Normal: *t-Student*

- **Definition**: Let $Z$ follow $N(0,1)$ and $Y_n$ follow $\chi^2_n$ then:

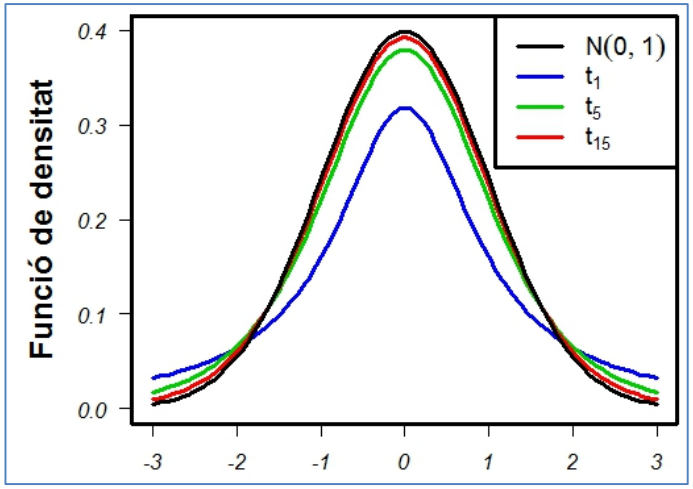
$$\frac{Z}{\sqrt{Y_n/n}} \sim t_n$$

[ when n→∞ (n>30), then $t_n \rightarrow N(0,1)$]

- **Notation**: $X \sim t_n$
- **Parameters**: *n* (degrees of freedom)
- **Density and Cumulative distribution function** :

$$f(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\cdot\Gamma(n/2)}\left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}} \quad for\ x>0$$

$$F(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n}\cdot B(1/2,n/2)}\left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}} \quad for\ x>0$$




*Γ*: Gamma function
*B*: Beta function
*n*: degrees of freedom

**R**: dt, pt, qt

*Script of a t-Student obtained from a Z and a Chi-square*

```
M = 500; n = 7                          # Sample length and degrees of freedom
samplez = rnorm(M, 0, 1)                # Sample of Normals
samplechi2 = rchisq(M,n)                # Sample of Chi-square
samplechi2n = sqrt(samplechi2/n)        # Denominator calculation
t = samplez / samplechi2n               # T-Student calculation
hist(t, breaks="Scott", freq=FALSE)     # Empirical distribution
curve(dt(x, n),add=TRUE,col=2,lwd=2)    # Theoretical distribution
quantile(t, c(0.25, 0.50, 0.75))        # Q1, Median and Q3 of Z/sqrt(Yn/n)
qt(c(0.25, 0.50, 0.75),n)               # Q1, Median and Q3 of Chi-square
```

# Distribution F Fisher-Snedecor

- **Definition**: Let $X_1$ follow $\chi^2_n$ and $X_2$ follow $\chi^2_m$ then:

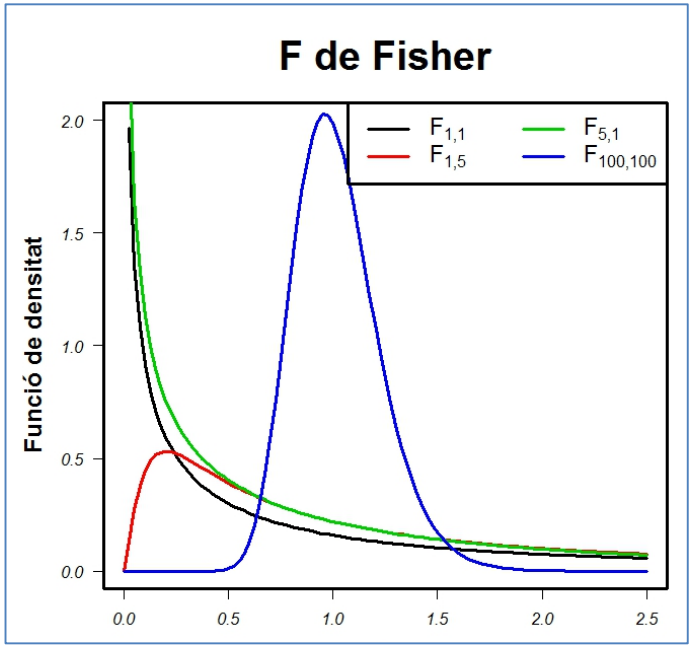
$$Y = \frac{X_1/n}{X_2/m} \sim F_{n,m}$$

*1/Y ~ $F_{m,n}$*

- **Notation**: $F \sim F_{n,m}$
- **Parameters**: *n* (numerator degrees of freedom)

  *m* (denominator degrees of freedom)
- **Density and Cumulative distribution function** :

[F distribution at Wikipedia]




F distribution will be used to compare two variances

**R**: df, pf, qf

*Script showing a F from a quotient of Chi-square*

```
M=500 ; n=5; m=7
samplechi2n = rchisq(M,n)
samplechi2m = rchisq(M,m)
F = (samplechi2n/n) / (samplechi2m/m)
hist(F, breaks="Scott", freq=FALSE)
curve(df(x, n, m),add=TRUE,col=2,lwd=2)
quantile(F, c(0.25, 0.50, 0.75))
qf(c(0.25, 0.50, 0.75), n, m)
```

# Probabilities and quantiles of RV models using R

```
# X is Bin(n=10,p=0.4)
dbinom(5,10,0.4)      # P(X=5)            → 0.2006581
pbinom(5,10,0.4)      # P(X<=5)           → 0.8337614
qbinom(0.5,10,0.4)    # P(X<=?)=0.5       → 4
# Y is Poi(lambda=4)
dpois(5,4)              # P(Y=5)          → 0.1562935
ppois(5,4)              # P(Y<=5)         → 0.7851304
qpois(0.5,4)            # P(Y<=?)=0.5     → 4
# E is exp(lambda=4)
pexp(1,4)               # P(E<=1)         → 0.9816844
qexp(0.5,4)           # P(E<=?)=0.5       → 0.1732868
# Z is N(0,1)
pnorm(1.96)             # P(Z<=1.96)      → 0.975
qnorm(0.95)           # P(Z<=?)=0.95 → ? = 1.645 = z0.95
```

```
# T is t15
pt(1.96,15)           # P(T<=1.96)        → 0.9655779
qt(0.95,15)           # P(T<=?)=0.95      → ? = 1.753 = t15,0.95
# X is Chi10
pchisq(5,10)          # P(X<=5)           → 0.108822
qchisq(0.95,10)       # P(X<=?)=0.95      → ? = 18.307
# F is F1,5
pf(1,1,5)             # P(F<=1)           → 0.6367825
qf(0.95,1,5)          # P(F<=?)=0.95      → ? = 6.607891
```

Function using R,
or online R:
https://rdrr.io/snippets/

```
# graphical normality analysis
#(30 random values
# from Normal          or Exponential)
x = rnorm(100)         # o x = rexp(100)
qqnorm(x)      # and then qqline(x)
```

